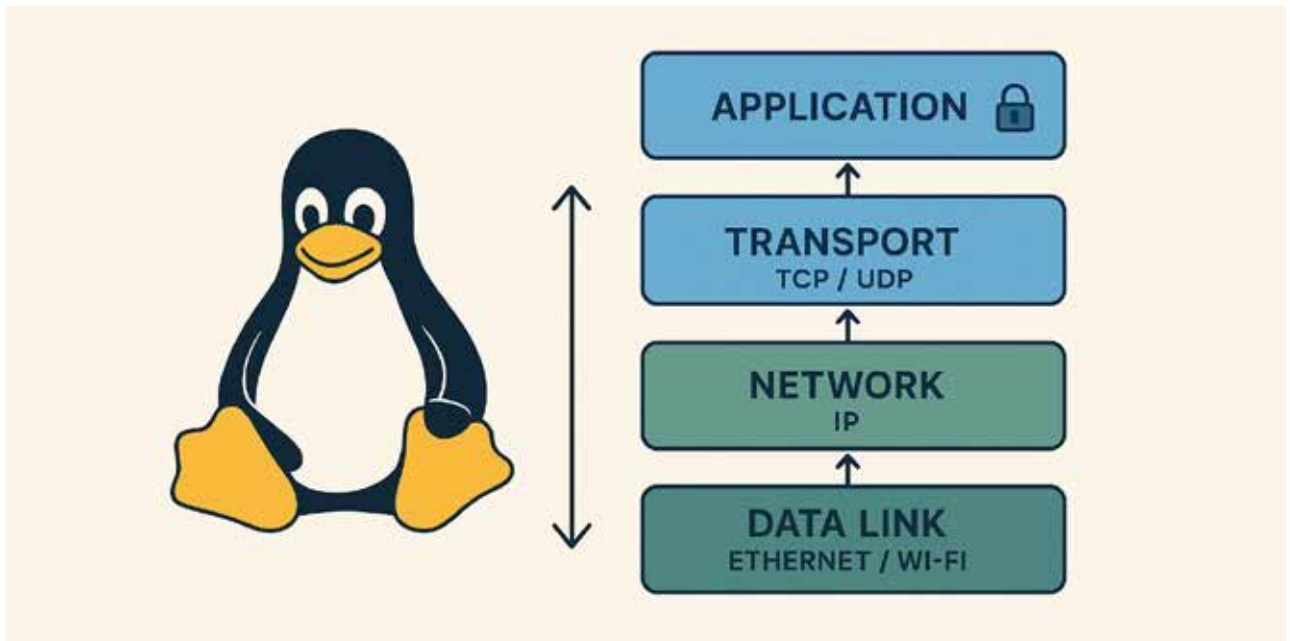


The Network Stack: Helping Linux Systems Communicate

The socket stack, the protocol stack and the network device drivers in the latest Linux versions offer great support for networking. This is how they work...



Version 0.01 of the Linux kernel was released to the public on May 14, 1991, with no support for networking. This first version relied heavily on 80386-compatible Intel processors and PC hardware. The release of Linux 1.0 on March 14, 1994, ushered a major revolution in open source technology as it introduced support for networking through UNIX's standard TCP/IP networking protocols as well as a BSD-compatible socket interface for network programming.

Ever since, Linux has undergone major upgrades and now has three layers — the socket stack, the protocol stack, and the network device drivers.

The socket stack: This layer takes care of all the networking requests by user applications.

The protocol stack: This is a framework for networking data arriving either from an application socket or a network device driver. It is tagged with an identifier, specifying which network

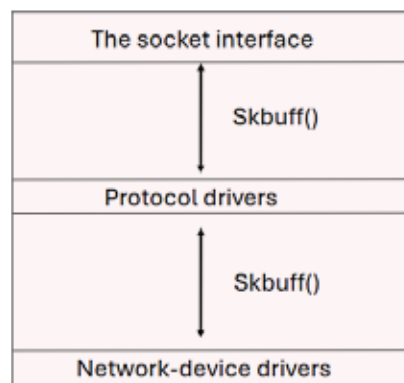


Figure 1: Communication between layers of the network stack

protocol they contain. Protocols can communicate with each other, manage routines, report errors and perform reliable transmission of lost data. The main task of this layer is to decide which socket or device it will send a packet to. It may rewrite the packets, split them, reassemble packets into fragments, or simply discard incoming data. Once the protocol layer has finished processing a set of packets, it passes them on upward to the socket interface or device driver according to the nature of the connection.

The communication between the layers of the network stack takes place by passing a single skbuff (socket buffer) structure. Each structure packs a set of pointers into a single contiguous memory representing a buffer,